Alessio Ferrarini

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https://alecsferra.github.io

Education

IMDEA Software Institute *PhD in Computer Science* Madrid, Spain 2024.09 – 2029 (expected)

- ⋆ Advisor: Prof. Niki Vazou
- ⋆ Research spans the metatheory of refinement type systems (soundness, consistency, and logical foundations) and their implementation in Liquid Haskell
- ⋆ Goal: Closing the gap between the expressiveness of dependent types and the automation of SMT-based refinement typing

University of Padua *Msc in Computer Science* Padua, Italy 2022.09 – 2024.07

- ⋆ 110/110 with honors - 29.9/30 GPA
- ⋆ Exchange student at VU Amsterdam (Amsterdam, Netherlands) - 9.3/10 GPA
- ⋆ Master thesis "Abstract Hoare Logic" advised by Prof. Francesco Ranzato and Prof. Paolo Baldan

University of Padua *BSc in Computer Science* Padua, Italy 2019.09 – 2022.07

- ⋆ 110/110 with honors - 28.8/30 GPA
- ⋆ Bachelor thesis "Implementation of a static typechecker and optimizations for a programming language" advised by Prof. Silvia Crafa

Publications

- ⋆ **CIRT: Calculus of Inductive Refinement Types**, *Under submission*
  Alessio Ferrarini, Niki Vazou, G.A. Kavvos. The first semantic foundation for refinement types with subtyping, impredicativity, SMT-style equality, stratified and positive inductive types.
- ⋆ **PLEX: Normalization for Refinement Types**, *OOPSLA 2026*
  Alessio Ferrarini, Niki Vazou, Wouter Swierstra. Extended Liquid Haskell's proof automation capabilities to automatically discharge proof obligations involving higher-order functions and type-level computation.
- ⋆ **Large Elimination and Indexed Types in Refinement Types (Extended abstract)**, *TYPES 2025*
  Alessio Ferrarini, Niki Vazou. Enabled dependent-type–style programming in refinement types by approximating indexed datatypes and large elimination.

Workshops

- ⋆ **Solid Code with Liquid Types!**, *Lambda World 2025*
  Pablo Castellanos, Alessio Ferrarini. Designed and delivered a 5-hour beginner to intermediate workshop on refinement types and formal verification using Liquid Haskell.

Work Experience

Amazon Web Services *Applied Scientist Intern* Minneapolis, MN, USA 2026.08 – 2026.11

- ⋆ Incoming Applied Scientist Intern at Amazon Web Services

University of Padua *Teaching Assistant* Padua, Italy 2022.09 – 2024.07

- ⋆ Teaching assistant for undergraduate courses: "Algorithms and Data Structures", "Automatons and Formal Languages", and "Programming 1"

Zucchetti S.p.A. *Intern* Padua, Italy 2022.05 – 2022.06

- ⋆ Developed type checker for the CPL programming language using C++ and Haskell
- ⋆ Improved runtime performance by 30% and implemented new language features

Accenture *Junior Developer* Verona, Italy 2019.07 – 2019.09

- ⋆ Developed an authentication system based on the OAuth2 specification using Java and TypeScript

Awards and Scholarships

- ⋆ *Mille e una Lode*, Recipient of "Mille e una Lode", an accolade bestowed upon the two best students in the Computer Science Master's program
- ⋆ *Cyber Challenge 2020*, Admitted to Cyber Challenge Italy (2020), a competitive national cybersecurity training program (acceptance rate: 12.57%)

Skills

- ⋆ *Natural Languages*: Italian (Native), Venetian (Native), English (Fluent), Spanish (Basic)
- ⋆ *Programming Languages*: (Liquid) Haskell, Agda, Lean, C++, Java, TypeScript, Rust, Python